

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

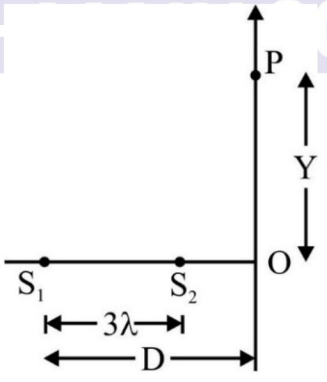
Sec: SR STAR MODEL-A & APEX, ALL IITS **DT: 09/01/2026**
Time: 3 Hrs **GTM - 13** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

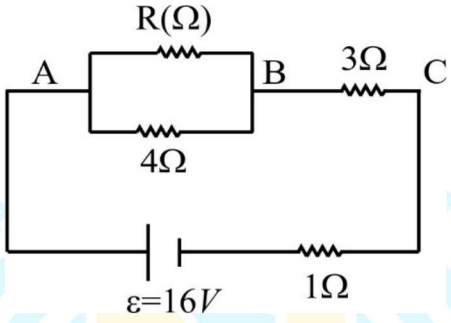
1. Two coherent narrow slits S_1, S_2 emitting light of wavelength λ in the same phase are placed parallel to each other at a small separation of 3λ. The light is collected on a screen S which is placed at a distance $D(>>\lambda)$ from the slits. The least distance Y from O such that the P is a maximum.



For More Join: @IITJEE_Advanc

A) $\sqrt{3}$D B) $\sqrt{8}$D C) $\sqrt{5}$D D) $\frac{\sqrt{5}\ D}{2}$

2. The potential difference between B and C is 9 volts. Then R value is



A) 3Ω B) 4Ω C) 1Ω D) 2Ω

3. E, M, J and G denote energy, mass, angular momentum and gravitational constant respectively. The dimensions of $\frac{EJ^2}{M^5G^2}$ are that of:

A) angle B) length C) mass D) time

4. The deflection in a moving coil galvanometer falls from 50 divisions to 10 divisions when a shunt of 12 ohm is applied. What is the resistance of the galvanometer?

A) 48Ω B) 46Ω C) 24Ω D) 12Ω

5. A particle A of mass m and initial velocity ϑ collides with a particle B of mass $\frac{m}{2}$ which is at rest. The collision is head on and elastic. The ratio of the de-Broglie wavelengths λ_A to λ_B after the collision is

A) $\frac{\lambda_A}{\lambda_B}=\frac{1}{3}$ B) $\frac{\lambda_A}{\lambda_B}=2$ C) $\frac{\lambda_A}{\lambda_B}=\frac{2}{3}$ D) $\frac{\lambda_A}{\lambda_B}=\frac{1}{2}$

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6. Read the two statements carefully to mark the correct option out of the options given below:

Statement-1: Smaller the orbit of the planet around the sun, shorter is the time it takes to complete one revolution.

Statement-2: According to Kepler's third law of planetary motion, square of time period is proportional to cube of mean distance from sun.

A) Statement-1 is true, statement-2 is true and Statement-2 is the correct explanation of Statement - 1.

B) Statement -1 is true, Statement-2 is true; Statement-2 is not correct explanation for Statement-1.

C) Statement-1 is true, Statement 2 is false.

D) Statement-1 is false, Statement 2 is true

7. A flask contains hydrogen and oxygen in the ratio of 2:1 by mass at temperature 27°C. The ratio of average kinetic energy per molecule of hydrogen and oxygen respectively is:

A) 2:1 B) 1:1 C) 1:4 D) 4:1

8. The excess pressure inside a spherical drop of water is three times that of another drop then their respective mass ratio is

A) 1:16 B) 8:1 C) 1:4 D) 1:27

9. If the distance between object and its two times magnified virtual image produced by a curved mirror is 15 cm, the focal length of the mirror must be:

A) $\frac{10}{3}$ cm B) – 12 cm C) – 10 cm D) 15 cm

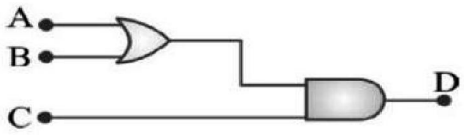
10. Two charges each of magnitude 0.01 C and separated by a distance of 0.4 mm constitute an electric dipole. If the dipole is placed in an uniform electric field ' $\vec{E}$ ' of 10 dyne/C making 30^0 angle with $\vec{E}$, the magnitude of torque acting on dipole is:

A) $4.0\times10^{-10}\,Nm$ B) $2.0\times10^{-10}\,Nm$

C) $1.0\times10^{-8}\,Nm$ D) $1.5\times10^{-9}\,Nm$

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11. To get an output 1 from the circuit shown in the figure, the input must be



A) A=0, B=1, C=0 B) A=1, B=0, C=0

C) A=1, B=0, C=1 D) A=1, B=1, C=0

12. In a hypothetical system, a particle of mass m and charge – 5q is moving around a very heavy particle of charge q. Assume that Bohr's model is applicable to this system, then the velocity of mass m in first Bohr's orbit is

A) $\frac{3q^2}{2\epsilon_0 h}$ B) $\frac{3q^2}{4\epsilon_0 h}$ C) $\frac{5q^2}{2\epsilon_0 h}$ D) $\frac{5q^2}{4\epsilon_0 h}$

13. For a transverse wave travelling along a straight line, the distance between two peaks (crests) is 5 m, while the distance between one crest and one trough is $1.5m$. The possible wavelengths (in m) of the waves are :

A) 1,3,5,….. B) $\frac{1}{1},\frac{1}{3},\frac{1}{5},....$ C) 1,2,3,….. D) $\frac{1}{2},\frac{1}{4},\frac{1}{6},....$

14. **Statement-1**: A parallel beam of light traveling in air can be displaced laterally by a parallel transparent slab by distance more than the thickness of the plate.

Statement-2: The lateral displacement of light travelling in air increases with rise in value of refractive index of slab.

A) both statements are true and Statement 2 is the correct explanation of Statement 1.

B) both statements are true but Statement 2 is not the correct explanation of Statement 1.

C) Statement 1 is true and Statement 2 is false.

D) Statement 1 is false but Statement 2 is true.

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15. **Statement I**: The ferromagnetic property depends on temperature. At high temperature, ferromagnet becomes paramagnet.

Statement II: At high temperature, the domain wall area of a ferromagnetic substance increases.

In the light of the above statements, choose the most appropriate answer from the options given below:

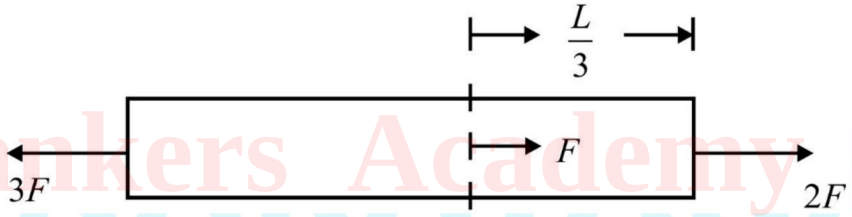
A) Statement I is true but Statement II is false

B) Both Statement I and Statement II are true

C) Both Statement I and Statement II are false

D) Statement I is false but Statement II is true

16. A uniform cylindrical rod of length L, cross-sectional area A and Young's modulus Y is acted upon by the forces as shown in the figure. The elongation of the rod is



A) $\frac{3FL}{5AY}$ B) $\frac{2FL}{5AY}$ C) $\frac{3FL}{8AY}$ D) $\frac{8FL}{3AY}$

17. An ideal gas at 27^0C is compressed adiabatically to $\frac{8}{27}$ of its original volume. If $\gamma = \frac{5}{3}$, then the rise in temperature of gas is

A) 450K B) 375 K C) 225 K D) 405 K

18. A helicopter is flying horizontally with a speed 'v' at an altitude 'h' has to drop a food packet for a man on the ground. What is the distance of helicopter from the man when the food packet is dropped?

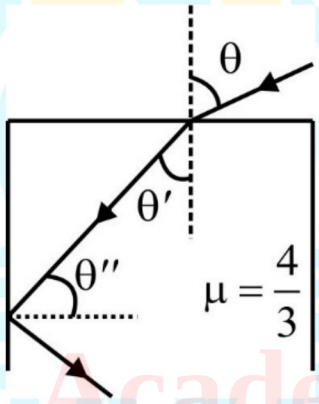
A) $\sqrt{\frac{2ghv^2+1}{h^2}}$ B) $\sqrt{2ghv^2+h^2}$ C) $\sqrt{\frac{2v^2h}{g}+h^2}$ D) $\sqrt{\frac{2gh}{v^2}+h^2}$

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19. A particle initially at rest starts moving from reference point x = 0 along x- axis, with velocity v that varies as $v = 4\sqrt{x}$ m/s. The acceleration of the particle in ms^{-2} is_____

A) 2 B) 4 C) 6 D) 8

20. Ray of light entering from air into a denser medium of refractive index $\frac{4}{3}$, as shown in figure. The light ray suffers total internal reflection at the adjacent surface as shown in the figure. The maximum value of angle θ should be equal to:



A) $\sin^{-1}\frac{\sqrt{7}}{3}$ B) $\sin^{-1}\frac{\sqrt{5}}{4}$ C) $\sin^{-1}\frac{\sqrt{7}}{4}$ D) $\sin^{-1}\frac{\sqrt{5}}{3}$

SECTION – II
(Numerical Value Answer Type)

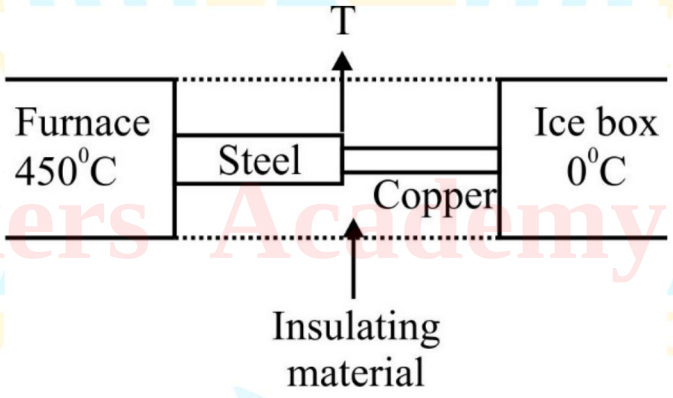
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. A parallel plate capacitor of capacitance 90 pF is connected to a battery of emf 20V. If dielectric material of dielectric constant $k = \frac{5}{3}$ is inserted between the plates, the magnitude of the induced charge will be $(0.3\times P)$ nano columbs. Find the value of P?

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22. In a series LR circuit, power of 400 W is dissipated from a source of 250 V,50 Hz. The power factor of the circuit is 0.8. In order to bring the power factor to unity, a capacitor of value C is added in series to the L and R. Taking the value C as $\left(\frac{n}{3\pi}\right)\mu F$, then value n is____

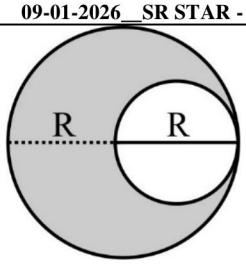
23. If K_1 and K_2 are the thermal conductivities, L_1 and L_2 are the lengths, A_1 and A_2 are the cross sectional areas of steel and copper rods respectively such that $\frac{K_2}{K_1}=9, \frac{A_1}{A_2}=2, \frac{L_1}{L_2}=2$ Then, for the arrangement as the shown in the figure, the value of temperature T of the steel copper junction in the steady state will be :



24. The electric field in a region is given by $\left(\frac{3}{5}E_0\hat{i}+\frac{4}{5}E_0\,\hat{j}\right)\frac{N}{C}$. The ratio of flux of reported field through the rectangular surface of area 0.2 m^2 (parallel to y-z plane) to that of the surface of area $0.3\,m^2$ (parallel to x-z plane) is a:b (in simplest form) where a =____

25. A uniform circular disc of radius 'R' and mass 'M' is rotating about an axis perpendicular to its plane and passing through its centre. A small circular part of radius R/2 is removed from the original disc as shown in the figure. The moment of inertia of the remaining part of the original disc about the axis as given above is $\frac{n}{32}MR^2$.Find the value of n ?

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Statement – I: Cysteine is an amino acid that contains a sulphur atom in its side chain, and it can form disulphide bonds to stabilize the protein structure.
Statement – II: Natural proteins are denatured by heating or by adding acids, which changes the primary structure of the proteins.
In the light of the above statements, choose the most appropriate answer from the options given below.
A) Both Statement – I and Statement – II are correct
B) Statement – I is correct and Statement – II is incorrect
C) Statement – I and Statement – II are incorrect
D) Statement – I is incorrect and Statement – II is correct

27. STATEMENT – I: Ammonia gas evolved by the action of $NaOH$ on Ammonium salts, reacts with HCl to give white dense fumes
STATEMENT – II: Addition of Nesseler's reagent to the aqueous solution of Ammonium salt in presence of KOH produces green precipitate of mercury (II) amido – Iodine.
A) Both Statement – I and Statement – II are correct
B) Statement – I is incorrect and Statement – II is correct
C) Statement – I is correct and Statement – II is incorrect
D) Both Statement – I and Statement – II are incorrect

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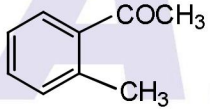
28. At 25^0C when 1 mole of $MgSO_4$ was dissolved in water, the heat evolved was found to be 80 kJ/mol. One mole of $MgSO_4.\ 7H_2O_{(s)}$ on dissolution gives a solution of the same composition accompanied by an absorption of 12 kJ/mol. The enthalpy of hydration for the reaction is $MgSO_{4(s)} + 7H_2O_{(l)} \rightarrow MgSO_4.7H_2O_{(s)}$

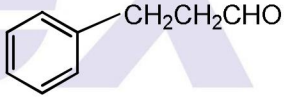
A) – 76 kJ/mol
B) +76 kJ/mol
C) +92 kJ/mol
D) – 92 kJ/mol

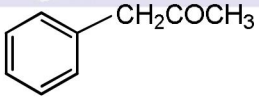
29. The decomposition of N_2O_5 takes place according to the following first order kinetics: $2N_2O_5(g) \rightarrow 2N_2O_4(g) + O_2(g)$, The maximum volume of O_2 gas obtained was 100 cm^3. If 90 cm^3 of O_2 gas is obtained in 500 minutes, then what is the rate constant (in min^{-1}) for disappearance of N_2O_5

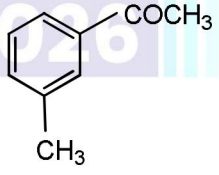
A) 2.303/500
B) 2.303/500 log 90/100
C) 500/4.606
D) 2.303/500 log 100/90

30. An organic compound **X** with molecular formula $C_9H_{10}O$ gives a positive test with **2,4-dinitrophenylhydrazine (2,4-DNP)**, and also gives a positive test with I_2 / OH^- (Iodoform test). Compound **X** does not react with **Tollen's reagent** or **Baeyer's reagent**. The compound **X** on oxidation with a strong oxidizing agent $K_2Cr_2O_7 / H^+$, followed by heat gives a carboxylic acid with molecular formula $C_7H_6O_2$. The compound **X** can be

A) 

B) 

C) 

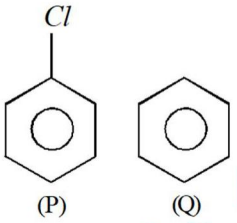
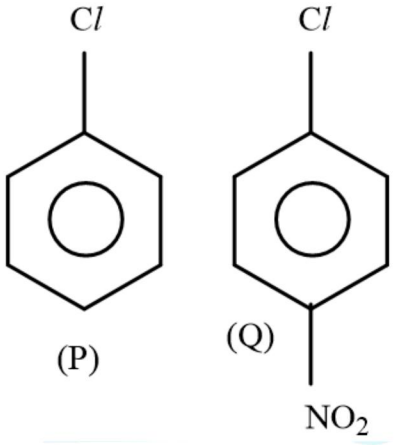
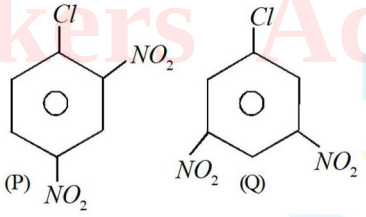
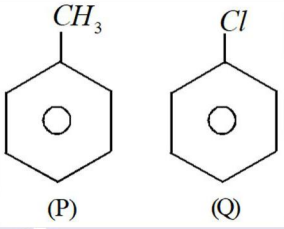
D) 

31. Which of the following is incorrect about the radii?

A) $I^- > I > I^+$
B) $Fe^{+3} < Fe^{+2} < Fe^+ < Fe$
C) $P^{-3} > S^{-2} > Ca^{+2} > Sc^{+3}$
D) P > S > Cl > Ar

For More Join: @IITJEE_Advanc

32. Match List – I with List – II

List – I		List – II	
1)		(I)	$P > Q$ towards electrophilic substitution reaction
2)		(II)	$Q > P$ towards electrophilic substitution reaction
3)		(III)	$P > Q$ towards nucleophilic substitution reaction
4)		(IV)	$Q > P$ towards nucleophilic substitution reaction

Choose the correct answer from the options given below

A) $1-II, 2-III, 3-IV, 4-I$

B) $1-II, 2-IV, 3-III, 4-I$

C) $1-I, 2-IV, 3-III, 4-II$

D) $1-I, 2-III, 3-IV, 4-II$

For More Join: @IITJEE_Advanc

33. Which of the following reactions defines heat of formation $\left(\Delta H_f^0\right)$?

A) C(diamond) + $O_2(g) \rightarrow CO_2(g)$

B) $\frac{1}{2}H_2(g)+\frac{1}{2}F_2(g) \rightarrow HF(g)$

C) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

D) $CO(g) + \frac{1}{2} O_2(g) \rightarrow CO_2(g)$

34. The vapour pressures of pure liquids A and B are 400 and 600 mm Hg respectively at 298 K. On mixing the two liquids, the sum of their partial volumes is equal to the volume of the final mixture. The mole fraction of liquid B is 0.5 in the mixture. The mole fractions of components A and B in vapour phase are:

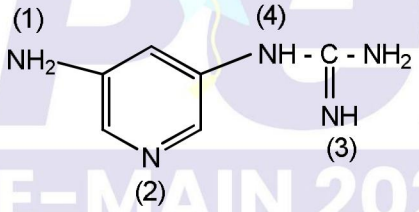
A) $3y_A = y_B$ B) $2y_A = 3y_B$ C) $3y_A = 2y_B$ D) $4y_A = 3y_B$

35. 'X' is the number of amphoteric oxides among the following oxides, $V_2O_3, Cr_2O_3, CrO, SnO_2, ZnO, Mn_2O_7, PbO_2$ and V_2O_5.

The primary valency of chromium in $\left[Cr(en)_3\right]_2(SO_4)_3$ is 'Y'. The value of $X+Y$ is

A) 6 B) 7 C) 8 D) 5

36. In the given structure arrange the nitrogen atoms in the decreasing order of their basic strength



Choose the correct answer from the options given below.

A) 1 > 2 > 3 > 4 B) 3 > 2 > 4 > 1

C) 3 > 2 > 1 > 4 D) 3 > 1 > 2 > 4

For More Join: @IITJEE_Advanc

37. Which of the following is incorrect statement

A) Mn_2O_7 is a covalent green oil

B) In the structure of covalent oxide Mn_2O_7 each $'Mn'$ is tetrahedrally surrounded by oxygen atoms including a $Mn-O-Mn$ bridge

C) The order of oxidizing power of $VO_2^+, Cr_2O_7^{-2}$ and MnO_4^- ions is $VO_2^+ > Cr_2O_7^- > MnO_4^-$

D) Potassium manganate disproportionates into potassium permanganate and manganese dioxide in acidic solution

38. Match List – I with List – II

List – I		List – II	
(1)	Solution of acetic acid in benzene	I	Minimum boiling azeotrope
(2)	Solution of chloroform & diethyl ether	II	$\Delta H_{mix} = 0$
(3)	Solution of Chlorobutane & Bromo butane	III	Dimerises
(4)	Solution of ethanol & acetone	IV	Maximum boiling azeotrope

Choose the correct answer from the options given below.

A) $(1)-II, (2)-III, (3)-II, (4)-IV$

B) $(1)-II, (2)-IV, (3)-I, (4)-III$

C) $(1)-III, (2)-IV, (3)-II, (4)-I$

D) $(1)-I, (2)-IV, (3)-III, (4)-II$

39. The hybridization of the nitrogen atom in ammonium nitrate (for both nitrogens), and sodium amide, respectively, are:

A) sp^3, sp^3, sp^3

B) sp^3, sp^2, sp^3

C) sp^2, sp^2, sp^3

D) sp^2, sp, sp^3

For More Join: @IITJEE_Advanc

40. The pair in which both species have same (spin only) magnetic moment is:

A) $[CrCl_4]^-$, $[FeCl_4]^-$ B) $[Ni(NH_3)_6]^{2+}$, $[Co(NH_3)_6]^{2+}$

C) $[Mn(Cl)_6]^{3-}$, $[FeF_6]^{3-}$ D) $[Mn(Cl)_6]^{3-}$, $[CoF_6]^{3-}$

41. An alkene (A) on reductive ozonolysis gave 1 mole of propanal, 1 mole of propanone, 1 mole of glyoxal and 2 moles of 2-methylpropane1,3-dial. Then the number of $\pi-$ bonds in the alkene (A) is____

A) 3 B) 4 C) 5 D) 6

42. Statement – I: Cis – 2 – butene is more polar than trans – 2 – butene

Statement – II: Boiling point of cis – 2 – butene is more than trans – 2 - butene

In the light of the above statements, choose the most \appropriate answer from the options given below:

A) Statement – I & Statement – II are correct

B) Statement – I is correct & Statement – II is incorrect

C) Statement – I & Statement – II are incorrect

D) Statement – I is incorrect & Statement – II is correct

43. Consider the following statements:

1) Symmetrical distribution of the electrons gives more stability for completely half-filled and full-filled orbitals.

2) d^5 configuration has a total possible number of exchanges are 10.

3) d^5 configuration has a more number of possible exchanges than d^{10} configuration because it has more number of unpaired electrons.

4) In writing the electronic configuration, 5s orbital is filled before 4p orbital.

Choose the incorrect statements

A) 3, 4 only B) 1, 3 & 4 only

C) 2 & 4 only D) 1, 2, 3 & 4 only

44. The correct statements from the following are:

1) The order of ionic size: $Ga^+ > In^+ > Tl^+$

2) The order of stability: $Tl^+ > In^+ > Ga^+$

3) The oxidation state of Thallium in TlI_3 is +3

4) $AlCl_3$ and $GaCl_3$ are covalent compounds when anhydrous.

For More Join: @IITJEE_Advanc

5) In the Boron family, Tl^{+3} has a more standard reduction potential and In^{+3} has least standard reduction potential.

Choose the correct answer from the options given below:

A) 3, 4 & 5 only

B) 1, 2, 3, 4 & 5

C) 2 & 4 only

D) 1, 3, 4 & 5 only

45. Given below are two statements:

For the reduction of NO_3^- ion from aqueous solution, E° is +0.96 V. Values of E° for some metal ions are listed below:

$V^{2+}(aq) + 2e^- \rightarrow V$: E° = –1.19 V

$Fe^{3+}(aq) + 3e^- \rightarrow Fe$: E° = – 0.04 V

$Au^{3+}(aq) + 3e^- \rightarrow Au$: E° = +1.40 V

$Hg^{2+}(aq) + 2e^- \rightarrow Hg$: E° = +0.86 V , based on the data given 2 statements

Statement – I: Fe & Hg are oxidised by NO_3^- in aq. solution

Statement – II: V & Au are not oxidised by NO_3^- in aq.solution.

In the light of the above statement, choose the most appropriate answer from the options given below:

A) Statement – I is correct & Statement – II is incorrect

B) Both Statement – I & Statement – II are incorrect

C) Both Statement – I & Statement – II are correct

D) Statement – I is incorrect & Statement – II is correct

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. One litre buffer solution was prepared by adding 0.1 mol each of CH_3COOH and CH_3COONa in deionized water. The change in pH on addition of 0.05 mol of $NaOH$ to the above solution is_____ $\times 10^{-2}$

(Given: P^{Ka} of $CH_3COOH = 4.7$ and $\log_{10} 3 = 0.48$)

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47. In Duma's method 200 mg of an organic compound released 49.2 ml of nitrogen gas at 300K temperature and 775 mm Hg pressure. The percentage composition of 'N' in the organic compound is _____ % (nearest integer)

(Aqueous tension at 300 K = 15 mm Hg) (R = 0.082 Lt. atm/mol. K)

48. $2A+4B \rightarrow 3C+4D$ (balanced reaction), when 5 moles of A react with 6 moles of B. The number of moles of C formed is $x\times10^{-1}$, the value of $'x'$ is_____

49. How many of the following statements are correct

1) $Hg\left[Co\left(SCN\right)_4\right]-$ Mercury (II) tetrathiocyanato – S – cobalt (II)

2) Square planar complex of the type M_{ABXL} (where A, B, X and L are unidentates) shows 3 isomers - two cis and one trans

3) $cis-\left[CrCl_2\left(OX\right)_2\right]^{-3}$ is optically active

4) $\left[CoF_6\right]^{-3}$ is outer orbital and high spin complex compound

5) Ligands for which $\Delta_0 < P$ are known as weak field ligands and form high spin complexes (Δ_0 = relative magnitude of the crystal field splitting, P=pairing energy)

6) $I^- < Cl^- < F^- < C_2O_4^{-2} < NH_3 < CN^-$ increasing order of ligand field strength

50. $\text{Phenol} \xrightarrow[\Delta]{Zn} X \xrightarrow[AlCl_3]{\text{Propanoyl chloride}} Y \xrightarrow{2.4-DNP} Z$

The number of sp^2 carbon atoms in the compound $'Z'$ is____

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If orthocenter of triangle formed by lines $x+3y=15, 2x-y-2=0$ and $x-y-1=0$ is H, then perpendicular distance of H from the line $x+y-3$ is

A) $\sqrt{2}$ B) $2\sqrt{2}$ C) $2\sqrt{3}$ D) $\sqrt{6}$

52. If $|\bar{a}|=1, |\bar{b}|=3, |\bar{c}|=4$ and $2\bar{a}+\bar{b}-\bar{c}=0$, then find the value of $|\bar{a}+2\bar{b}|$

A) $2\sqrt{10}$ B) $3\sqrt{10}$ C) $4\sqrt{2}$ D) $6\sqrt{2}$

53. Let set $A=\{x\,||x-1|\le 2\}$

Set $B=\{y\,|\,2\le|y+1|<6\}$

Set $C=A-B$ then C is

A) $[-1,1]$ B) $(-1,1)$ C) $(-1,1]$ D) $[-1,1)$

54. The range of value of λ for which the expression $\frac{2x^2-5x+3}{4x-\lambda}$ can take all real values for $x\in R-\left\{\frac{\lambda}{4}\right\}$, is:

A) (4,6) B) $[4,6]$ C) $(4,6]$ D) $[4,6)$

55. A bag contains 8 unbiased coins and 2 biased coins with probability of tail is $\frac{1}{3}$.

If one coin is drawn from bag and found to be head what is the probability that drawn coin is biased coin.

A) $\frac{1}{4}$ B) $\frac{1}{6}$ C) $\frac{1}{3}$ D) $\frac{2}{3}$

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56. Let X be a random variable, and let $P(X=x)$ denote the probability that X takes the values x. Suppose that the points $(x, P(X=x))$, $x=0,1,2,3,4$, lie on a fixed straight line in the $xy-$plane, and $P(X=x)=0$ for all $x \in R-\{0,1,2,3,4\}$. If the mean of X is $\frac{5}{2}$ and the variance of X is α, then the value of 24α is.

A) 36 B) 42 C) 48 D) 72

57. If the area of the region $\left\{(x,y) \in R^2 : y^2 \geq 4x, |y| \leq \frac{x}{2}+2\right\}$ is Δ, then find the value of $[\Delta]$. [S] denotes greatest integer less than or equal to S.]

A) 7 B) 8 C) 9 D) 10

58. Let $f: R \to R$ be a polynomial function of degree 4 having extreme value at $x=\frac{1}{8}$. If $\lim\limits_{x\to 0}\frac{f(x)}{x^3}=1$, then $\lim\limits_{x\to 0}\frac{f(x)-x^3}{x^4}$ is

A) 3 B) 6 C) -3 D) -6

59. Number of solutions of the trigonometric equation $\sin 3\theta = 4\sin\theta \sin 2\theta \sin 4\theta$ in $[0,\pi]$ is

A) 4 B) 6 C) 8 D) 10

60. Let T_n be n^{th} term of an A.P. If $T_6+T_{10}+T_{19}+T_{12}+T_{13}+T_{15}=600$ then sum of first 24 terms of A.P. is

A) 1200 B) 1800 C) 2400 D) 3600

61. Let z be a complex number with non-zero imaginary part. If $\frac{2+3z+4z^2}{2-3z+4z^2}$ is a real number, then the value of $|Z|^2$ is

A) $\frac{1}{2}$ B) $\frac{1}{3}$ C) 1 D) 2

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62. On a coordinate plane ellipse $C_1:\frac{x^2}{a_1^2}+\frac{y^2}{b_1^2}=1(a_1>b_1>0)$ and hyperbola $C_2:\frac{x^2}{a_2^2}-\frac{y^2}{b_2^2}=1(a_2>b_2>0)$ has the same focus point F_1,F_2. Point P is the intersection point of C_1 and C_2 in the first quadrant and $|F_1F_2|=2|PF_2|$. e_1 is the eccentricity of C_1 and e_2 is the eccentricity of C_2. Then range of e_2-e_1 is

A) $\left(\frac{1}{3},\infty\right)$ B) $\left(\frac{1}{2},\infty\right)$ C) $\left[\frac{1}{3},\infty\right)$ D) $\left[\frac{1}{2},\infty\right)$

63. A function $y=f(x)$ satisfies $x\,f'(x)-2f(x)=x^4f^2(x),\forall x>0$ and $f(1)=-6$ then the value of $f'(3^{1/5})$ is

A) 6 B) 8 C) 9 D) 27

64. A line l passing through the origin is perpendicular to the lines

$l_1:(3+t)\hat{i}+(-1+2t)\hat{j}+(4+2t)\hat{k},\ -\infty<t<\infty,$

$l_2:(3+2s)\hat{i}+(3+2s)\hat{j}+(2+s)\hat{k},\ -\infty<s<\infty.$

Then, one of the coordinate of the points on l_2 at a distance of $\sqrt{17}$ from the point of intersection l and l_1 is

A) $\left(\frac{7}{3},\frac{7}{3},\frac{5}{3}\right)$ B) (3,3,2) C) (1,1,1) D) $\left(\frac{7}{9},\frac{7}{9},\frac{8}{9}\right)$

65. The number of quadrilaterals which can be formed using vertices of decagon such that no side of quadrilateral coincide with the side of decagon are

A) 20 B) 25 C) 30 D) 45

66. If the area of triangle formed by $A(1,2,0),B(2,3,-1)$ and foot of perpendicular drawn from B on the line $L\equiv x-2=\frac{y-1}{2}=1-z$ is S, then $2S^2$ is____

A) 2 B) 3 C) 4 D) 8

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67. Solution set of the equation $\sqrt{4^x - 2^{x+1} + 1} + \sqrt{4^x - 2^{x+3} + 16} = 3$ is:

A) $x \in (0,2)$ B) $x \in (0,2]$ C) $x \in [0,2]$ D) $x \in = \{0,2\}$

68. Two newspapers A and B are published in a city. It is known that 25% of the city population reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisements is_____

A) 12.8 B) 13 C) 13.5 D) 13.9

69. If the following system of linear equations

$2x + y + z = 5$

$x - y + z = 3$

$x + y + az = b$

have no solutions then

A) $a = -\frac{1}{3}, b \neq \frac{7}{3}$ B) $a \neq \frac{1}{3}, b = \frac{7}{3}$

C) $a \neq -\frac{1}{3}, b = \frac{7}{3}$ D) $a = \frac{1}{3}, b \neq \frac{7}{3}$

70. Let $x_1, x_2, ...x_{10}$ be ten observations such that $\sum_{i=1}^{10}(x_i - 2) = 30, \sum_{i=1}^{10}(x_i - \beta)^2 = 98$, $\beta > 2$ and their variance is $\frac{4}{5}$. If μ *and* σ^2 are respectively the mean and the variance of $2(x_1 - 1) + 4\beta, 2(x_2 - 1) + 4\beta, ..., 2(x_{10} - 1) + 4\beta$, then $\frac{\beta\mu}{\sigma^2}$ is equal to

A) 100 B) 110 C) 120 D) 90

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. If $\lim\limits_{x\to 0}\dfrac{\cos(\tan x)-\cos x}{x^4}$ is equal to K. Find the value of $18K^2$

72. If the value of the integral

$$\int \frac{\sin\theta.\sin 2\theta\left(\sin^6\theta+\sin^4\theta+\sin^2\theta\right)\sqrt{2\sin^4\theta+3\sin^2\theta+6}}{1-\cos 2\theta}d\theta \text{ is}$$

$$\frac{\left(a\sin^6\theta+b\sin^4\theta+c\sin^2\theta\right)^{3/2}}{d}+K\,.\,(a,b,c,d\in N \text{ and}$$

$H.C.F.(a,b)=1)$ and K is integral constant. Then the value of $a+b+c+d$ is

73. A bag contains 6 white and 4 black balls. A die is rolled once and the number of balls equal to the number obtained on the die are drawn from the bag at random. The probability that all the balls drawn are white is $\dfrac{m}{n}$, where m,n are co-primes, then m^2+n^2 is_____

74. An ellipse and a hyperbola have their principal axes along the coordinate axes and have common foci separated by a distance $2\sqrt{13}$. The difference of their focal semi-axes is equal to 4. If the ratio of their eccentricities is $\dfrac{3}{7}$, if the equation of ellipse is $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ and equation of hyperbola is $\dfrac{x^2}{c^2}-\dfrac{y^2}{d^2}=1$ then the value of $a^2-b^2+c^2-d^2$ is

75. The value of $C_0-2^2C_1+3^2C_2-4^2C_3+....-20^2C_{19}$ is_____ (where $C_r={}^{19}C_r$)

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